

Clinical Trial Protocol

Reconstructed from CDISCPilot01 Source Data

mrgda example-project

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1 Administrative Information

This protocol is reconstructed from the XPT source data in `inst/example-project/data/source/CDISCPIL0T01`. It is intended for example-project documentation and data interpretation. It is not an original sponsor-approved protocol and must not be used to conduct human subject research.

Study identifier: CDISCPIL0T01

Protocol title: A randomized, placebo-controlled, parallel-group study of transdermal Xanomeline in subjects with mild to moderate Alzheimer’s disease.

Investigational product: Xanomeline transdermal patch.

Control: Matching placebo transdermal patch.

Country and sites: USA; 17 sites.

Observed subject reference period: 2012-07-09 to 2015-03-05.

Observed exposure period: 2012-07-09 to 2015-03-05.

2 Protocol Synopsis

Item	Description
Design	Multicenter, randomized, placebo-controlled, parallel-group clinical pharmacology and safety study.
Population	Adult and older adult subjects with Alzheimer’s disease, inferred from the CDISC pilot Xanomeline study context and treatment program.
Sample size in source data	306 subjects in demographics, including 254 non-screen-failure records and 254 records with non-screen-failure actual treatment.
Treatment groups	Placebo; Screen Failure; Xanomeline High Dose; Xanomeline Low Dose
Route and dosage form	Once-daily transdermal patch.
Primary objective	To evaluate safety, tolerability, and pharmacokinetic exposure for Xanomeline low-dose and high-dose patches compared with placebo.
Key assessments	Exposure, laboratory tests, pharmacokinetic concentrations, and vital signs.
PK sampling	Plasma and urine Xanomeline concentrations at baseline from pre-dose through 48 hours post-dose.
Key scheduled visits	SCREENING 1 (Day -7); SCREENING 2 (Day -1); BASELINE (Day 1); AMBUL ECG PLACEMENT (Day 13); WEEK 2 (Day 14); WEEK 4 (Day 28); AMBUL ECG REMOVAL (Day 30); WEEK 6 (Day 42); WEEK 8 (Day 56); WEEK 12 (Day 84); WEEK 16 (Day 112); WEEK 20 (Day 140); WEEK 24 (Day 168); WEEK 26 (Day 182); RETRIEVAL (Day 168)

3 Background and Rationale

The CDISCPIL0T01 source package contains transdermal Xanomeline exposure records, laboratory results, pharmacokinetic concentrations, and vital signs. The treatment arms include placebo, Xanomeline low dose, and Xanomeline high dose. This reconstructed protocol therefore describes a placebo-controlled Xanomeline study with primary emphasis on safety, tolerability, and pharmacokinetics.

The available source package does not include efficacy, adverse event, disposition, concomitant medication, or medical history domains. Those omissions limit reconstruction of the original clinical rationale and endpoint hierarchy. The protocol below uses the available source data as the controlling evidence and flags inferred content where appropriate.

4 Objectives and Endpoints

4.1 Primary Objective

To characterize the safety and tolerability of once-daily transdermal Xanomeline low-dose and high-dose patches compared with placebo.

4.2 Secondary Objectives

1. To characterize plasma and urine Xanomeline concentration-time profiles after dosing.
2. To summarize laboratory safety findings by category and reference range indicator.
3. To summarize vital sign findings over scheduled and unscheduled visits.
4. To summarize exposure to placebo, Xanomeline 54 mg, and Xanomeline 81 mg transdermal patches.

4.3 Endpoints

Primary safety endpoints include laboratory reference range indicators, vital sign values, deaths recorded in demographics, and exposure interruptions or dosing patterns where available. Pharmacokinetic endpoints include observed Xanomeline concentration by matrix, nominal time point, and visit.

Because an adverse event domain is not present in this source package, treatment-emergent adverse event incidence cannot be reconstructed from these files.

5 Study Design

This is reconstructed as a multicenter, randomized, placebo-controlled, parallel-group study. Subjects were screened, randomized to placebo or one of two Xanomeline dose levels, and followed through Week 24 with additional Week 26 or retrieval records represented in vital signs. Exposure records show dosing at baseline, Week 2, and Week 24.

Table 1: Planned Treatment Arms

Planned Arm	Subjects
Placebo	86
Screen Failure	52
Xanomeline High Dose	84
Xanomeline Low Dose	84

Table 2: Actual Treatment Arms

Actual Arm	Subjects
Placebo	86
Screen Failure	52
Xanomeline High Dose	72
Xanomeline Low Dose	96

Table 3: Subjects by Site

Site	Subjects
701	51
710	38
708	32
716	29
704	25
709	23
705	21
703	19
718	13
711	12
715	12
713	9
717	7
714	6
707	5
706	3
702	1

6 Study Population

6.1 Inclusion Criteria

Based on the source data and the CDISC pilot Xanomeline context, eligible subjects would have met all of the following criteria:

1. Adult or older adult male or female subjects.
2. Clinical diagnosis consistent with Alzheimer’s disease.
3. Ability to receive a transdermal patch and complete protocol-required safety and PK assessments.
4. Availability for follow-up through Week 24.
5. Written informed consent provided by the subject or legally authorized representative.

6.2 Exclusion Criteria

Subjects would have been excluded for any of the following:

1. Clinically significant unstable medical condition that could confound safety assessment.
2. Screening laboratory or vital sign finding considered clinically unacceptable by the investigator.
3. Known hypersensitivity or contraindication to Xanomeline or patch components.
4. Use of prohibited concomitant medications or participation in another interventional study.
5. Any condition preventing reliable PK sampling or follow-up.

6.3 Demographic Characteristics in Source Data

Table 4: Sex

Characteristic	Value
Subjects	306

Characteristic	Value
Non-screen-failure planned treatment records	254
Non-screen-failure actual treatment records	254
Age, mean	75.1
Age, median	77
Age, range	50-89 years
Female	179
Male	127
Deaths flagged in demographics	3

Table 5: Race

Sex	Subjects
F	179
M	127

Table 6: Ethnicity

Race	Subjects
WHITE	273
BLACK OR AFRICAN AMERICAN	29
AMERICAN INDIAN OR ALASKA NATIVE	2
ASIAN	2

Ethnicity	Subjects
HISPANIC OR LATINO	17
NOT HISPANIC OR LATINO	289

7 Investigational Product and Administration

The investigational product is Xanomeline administered as a once-daily transdermal patch. Exposure records include placebo patches and active Xanomeline patches at 54 mg and 81 mg.

Table 7: Exposure Regimens

	Treatment	Dose	Unit	Form	Frequency	Route
1	PLACEBO	0	mg	PATCH	QD	TRANSDERMAL
6	XANOMELINE	54	mg	PATCH	QD	TRANSDERMAL
7	XANOMELINE	81	mg	PATCH	QD	TRANSDERMAL

Table 8: Exposure Visits

Visit Number	Visit	Study Day
3	BASELINE	1
4	WEEK 2	14
12	WEEK 24	168

Treatment compliance, patch application site rotation, accountability, and dose modification rules are not represented in the source domains and would need to be supplied by the original protocol or operational manuals.

8 Schedule of Assessments

The following visits are represented across the source domains. Unscheduled visits are retained because the laboratory and vital sign data include them.

Table 9: Visits Represented in Source Data

	Visit Number	Visit	Study Day
1	1.0	SCREENING 1	-7
5090	1.1	UNSCHEDULED 1.1	NA
10969	1.2	UNSCHEDULED 1.2	NA
10851	1.3	UNSCHEDULED 1.3	NA
64156	2.0	SCREENING 2	-1
9857	3.0	BASELINE	1
88473	3.1	UNSCHEDULED 3.1	NA
48760	3.5	AMBUL ECG PLACEMENT	13
2	4.0	WEEK 2	14
6728	4.1	UNSCHEDULED 4.1	NA
20007	4.2	UNSCHEDULED 4.2	NA
3	5.0	WEEK 4	28
1685	5.1	UNSCHEDULED 5.1	NA
1173	6.0	AMBUL ECG REMOVAL	30
1159	6.1	UNSCHEDULED 6.1	NA
4	7.0	WEEK 6	42
37350	7.1	UNSCHEDULED 7.1	NA
5	8.0	WEEK 8	56
21694	8.2	UNSCHEDULED 8.2	NA
6	9.0	WEEK 12	84
3385	9.2	UNSCHEDULED 9.2	NA
3386	9.3	UNSCHEDULED 9.3	NA
7	10.0	WEEK 16	112
8	11.0	WEEK 20	140
9	12.0	WEEK 24	168
28226	12.1	UNSCHEDULED 12.1	NA
10	13.0	WEEK 26	182
56257	13.1	UNSCHEDULED 13.1	NA
46973	201.0	RETRIEVAL	168

Table 10: Reconstructed Schedule of Assessments

Assessment	Screening	Baseline	Weeks 2-24	Week 26/Retrieval	Unscheduled
Demographics	X				
Exposure		X	X		
Laboratory tests	X	X	X		X
Pharmacokinetic concentrations		X			
Vital signs	X	X	X	X	X

9 Safety Assessments

9.1 Laboratory Assessments

Laboratory testing includes chemistry, hematology, urinalysis, and additional miscellaneous results. The source package contains 59580 laboratory records.

Table 11: Laboratory Results by Category

Category	Results
CHEMISTRY	32740
HEMATOLOGY	21919
URINALYSIS	4370
OTHER	543
Not reported	8

Table 12: Laboratory Results by Reference Range Indicator

Reference Range Indicator	Results
NORMAL	56855
HIGH	1538
LOW	864
ABNORMAL	318
Not reported	5

Table 13: Laboratory Tests

	Category	Code	Test	Unit
1	CHEMISTRY	ALB	Albumin	g/L
11	CHEMISTRY	ALP	Alkaline Phosphatase	U/L
21	CHEMISTRY	ALT	Alanine Aminotransferase	U/L
32	CHEMISTRY	AST	Aspartate Aminotransferase	U/L
50	CHEMISTRY	BILI	Bilirubin	umol/L
60	CHEMISTRY	BUN	Blood Urea Nitrogen	mmol/L
70	CHEMISTRY	CA	Calcium	mmol/L
80	CHEMISTRY	CHOL	Cholesterol	mmol/L
90	CHEMISTRY	CK	Creatine Kinase	U/L
100	CHEMISTRY	CL	Chloride	mmol/L
114	CHEMISTRY	CREAT	Creatinine	umol/L
132	CHEMISTRY	GGT	Gamma Glutamyl Transferase	U/L
142	CHEMISTRY	GLUC	Glucose	mmol/L
172	CHEMISTRY	K	Potassium	mmol/L
246	CHEMISTRY	PHOS	Phosphate	mmol/L
266	CHEMISTRY	PROT	Protein	g/L
232	CHEMISTRY	SODIUM	Sodium	mmol/L
291	CHEMISTRY	URATE	Urate	umol/L
31	HEMATOLOGY	ANISO	Anisocytes	Not reported
42	HEMATOLOGY	BASO	Basophils	GI/L
59533	HEMATOLOGY	BASOLE	Basophils/Leukocytes	FRACTION
124	HEMATOLOGY	EOS	Eosinophils	GI/L
59535	HEMATOLOGY	EOSLE	Eosinophils/Leukocytes	FRACTION
152	HEMATOLOGY	HCT	Hematocrit	l
162	HEMATOLOGY	HGB	Hemoglobin	mmol/L
186	HEMATOLOGY	LYM	Lymphocytes	GI/L
59537	HEMATOLOGY	LYMLE	Lymphocytes/Leukocytes	FRACTION
609	HEMATOLOGY	MACROCY	Macrocytes	Not reported
194	HEMATOLOGY	MCH	Ery. Mean Corpuscular Hemoglobin	fmol(Fe)
204	HEMATOLOGY	MCHC	Ery. Mean Corpuscular HGB Concentration	mmol/L
214	HEMATOLOGY	MCV	Ery. Mean Corpuscular Volume	fL

	Category	Code	Test	Unit
25555	HEMATOLOGY	MICROCY	Microcytes	Not reported
224	HEMATOLOGY	MONO	Monocytes	GI/L
59539	HEMATOLOGY	MONOLE	Monocytes/Leukocytes	FRACTION
256	HEMATOLOGY	PLAT	Platelet	GI/L
19390	HEMATOLOGY	POIKILO	Poikilocytes	Not reported
684	HEMATOLOGY	POLYCHR	Polychromasia	Not reported
276	HEMATOLOGY	RBC	Erythrocytes	TI/L
306	HEMATOLOGY	WBC	Leukocytes	GI/L
15200	Not reported	HBA1C	Hemoglobin A1C	1
290	OTHER	TSH	Thyrotropin	mU/L
305	OTHER	VITB12	Vitamin B12	pmol/L
110	URINALYSIS	COLOR	Color	Not reported
182	URINALYSIS	KETONES	Ketones	Not reported
242	URINALYSIS	PH	pH	Not reported
286	URINALYSIS	SPGRAV	Specific Gravity	Not reported
301	URINALYSIS	UROBIL	Urobilinogen	Not reported

9.2 Vital Signs

Vital signs include blood pressure, pulse rate, height, temperature, and weight. Blood pressure and pulse were collected in supine and standing positions where applicable.

Table 14: Vital Sign Parameters

	Code	Vital Sign	Position	Location	Unit
2	DIABP	Diastolic Blood Pressure	STANDING	Not reported	mmHg
4965	DIABP	Diastolic Blood Pressure	STANDING	Not reported	Not reported
1	DIABP	Diastolic Blood Pressure	SUPINE	Not reported	mmHg
43	HEIGHT	Height	Not reported	Not reported	cm
45	PULSE	Pulse Rate	STANDING	Not reported	BEATS/MIN
4996	PULSE	Pulse Rate	STANDING	Not reported	Not reported
44	PULSE	Pulse Rate	SUPINE	Not reported	BEATS/MIN
21787	PULSE	Pulse Rate	SUPINE	Not reported	Not reported
87	SYSBP	Systolic Blood Pressure	STANDING	Not reported	mmHg
5026	SYSBP	Systolic Blood Pressure	STANDING	Not reported	Not reported
86	SYSBP	Systolic Blood Pressure	SUPINE	Not reported	mmHg
21811	SYSBP	Systolic Blood Pressure	SUPINE	Not reported	Not reported
12138	TEMP	Temperature	Not reported	EAR	C
128	TEMP	Temperature	Not reported	ORAL CAVITY	C
142	WEIGHT	Weight	Not reported	Not reported	kg

9.3 Deaths

The demographics domain contains 3 subjects flagged with death. The source package available here does not contain adverse event, disposition, or death-detail domains, so causality, timing relative to treatment, and event narratives cannot be reconstructed from these files.

10 Pharmacokinetic Assessments

The pharmacokinetic concentration domain contains 4572 records for Xanomeline in plasma and urine. Sampling is represented at baseline from pre-dose through 48 hours post-dose, including interval collections for urine.

Table 15: PK Analytes and Specimens

	Code	Analyte	Unit	Specimen	Lower Limit
1	XAN	XANOMELINE	ug/ml	PLASMA	0.01
15	XAN	XANOMELINE	ug/ml	URINE	0.01

Table 16: PK Nominal Time Points

	Visit	Study Day	Nominal Time Point	Nominal Hour
1	BASELINE	1	Pre-dose	-0.50
2	BASELINE	1	5 Min Post-dose	0.08
3	BASELINE	1	30 Min Post-dose	0.50
4	BASELINE	1	1h Post-dose	1.00
5	BASELINE	1	1.5h Post-dose	1.50
6	BASELINE	1	2h Post-dose	2.00
15	BASELINE	1	0-6h Post-dose	3.00
7	BASELINE	1	4h Post-dose	4.00
8	BASELINE	1	6h Post-dose	6.00
9	BASELINE	1	8h Post-dose	8.00
16	BASELINE	1	6-12h Post-dose	9.00
10	BASELINE	1	12h Post-dose	12.00
11	BASELINE	1	16h Post-dose	16.00
17	BASELINE	1	12-24h Post-dose	18.00
12	BASELINE	1	24h Post-dose	24.00
13	BASELINE	1	36h Post-dose	36.00
18	BASELINE	1	24-48h Post-dose	37.00
14	BASELINE	1	48h Post-dose	48.00

PK analyses should summarize concentration-time data by treatment, specimen, and nominal time point. Values below the lower limit of quantification should be handled using predefined rules before noncompartmental analysis or descriptive summaries.

11 Statistical Considerations

All analyses for this reconstructed protocol should be descriptive. The safety analysis set should include all subjects with non-screen-failure actual treatment assignment, using the demographics actual arm field as the available proxy for treatment received. PK summaries should include subjects with at least one quantifiable or reportable PK concentration record.

Continuous variables should be summarized with n, mean, standard deviation, median, minimum, and maximum. Categorical variables should be summarized with counts and percentages. Laboratory and vital sign summaries should be presented by visit and treatment arm. PK concentrations should be summarized by specimen, nominal time point, and actual treatment arm.

No formal hypothesis testing is specified because this reconstruction is based on source domains rather than an original statistical analysis plan.

12 Data Handling and Quality Control

The source package should be checked for:

1. Consistency of STUDYID, USUBJID, site, arm, and actual arm values across domains.

2. Consistency of visit numbers, visit names, and study days.
3. Completeness of exposure records for treated subjects.
4. Validity of laboratory reference range indicators and standardized units.
5. Consistency of PK time point labels and nominal time values.
6. Traceability from generated outputs to the source XPT records.

13 Ethical and Regulatory Considerations

For an actual trial, the protocol and informed consent materials would require institutional review board or ethics committee approval before initiation. The study would be conducted in accordance with Good Clinical Practice, applicable regulations, and the Declaration of Helsinki.

14 Limitations of Reconstruction

This document is inferred from available XPT source data. The source package does not include the original protocol, adverse event data, disposition data, eligibility data, concomitant medication data, efficacy assessments, randomization metadata, or study drug accountability details. Those records should be incorporated from source systems or original study documents before using this as anything beyond an example protocol.